

Initial Report - Group 4

Chosen legacy code

Request, an HTTP client library for Node.js.

Source: <https://github.com/request/request>

The owners of request officially deprecated it in February 2020, and the last commit on master was made on dated 2020-02-11. It is therefore defined as a legacy system by the owners and not something argued by us. It holds 2,270 commits between 2010 and 2020, and is made up of 2,991 lines of library source with 8,457 lines of tests across 55 files (1,320 assertions).

We required that a potential codebase could clone, build and run its test suite on the first attempt. Without a working baseline, no modification can be shown to improve anything.

Programming language

JavaScript on Node.js.

Type of maintenance planned

Adaptive (ISO/IEC/IEEE 14764) by definition, since it is done on legacy code.

The tests in the repository fail because of the SSL certificates having been created for an older version of OpenSSL, which means the environment has jumped since the last working version. The code hasn't changed, only the environment has, which places this under adaptive maintenance.

Preliminary plan

In general the plan is to work in a "Cover and Modify" approach and also with Feathers' Methodology throughout the project. The more specific phases considered would be:

Phase 1: Investigate the codebase to get a working build, and get the tests running properly.

Phase 2: Find out the root cause of test failures.

Phase 3: Fix the identified failures and verify that no tests have regressed.

Phase 4: Document the fixes and findings, map it to the chosen maintenance type, and work on the presentation.